

ORIGINAL ARTICLE

Ten-Year Outcomes after CAR T-Cell Therapy for B-Cell Lymphomas

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ABSTRACT

BACKGROUND

Anti-CD19 chimeric antigen receptor (CAR) T-cell therapy is a standard treatment for relapsed or refractory B-cell non-Hodgkin lymphomas. Long-term results and curative potential remain uncertain.

METHODS

We evaluated long-term outcomes in 38 patients with relapsed or refractory B-cell non-Hodgkin lymphomas (24 patients with large B-cell lymphoma and 14 with follicular lymphoma) who had been treated with CTL019 (now called tisagenlecleucel) — autologous T cells expressing CD19-directed, 4-1BB–costimulated chimeric receptors. Lymphoma-free survival was defined as the time from the tisagenlecleucel infusion to relapse or lymphoma-related death. The incidence of non–relapse-related death and second primary cancer was estimated with the Aalen–Johansen method. The data-cutoff date was October 1, 2025.

RESULTS

At a median follow-up of 10.1 years (range, 7.9 to 11.5), no relapses had occurred beyond 5.4 years. The 10-year lymphoma-free survival was 32% (95% confidence interval [CI], 14 to 51) among patients with large B-cell lymphoma and 47% (95% CI, 20 to 71) among those with follicular lymphoma. In an analysis that included deaths from any cause, the 10-year progression-free survival was 17% (95% CI, 5 to 34) among patients with large B-cell lymphoma and 29% (95% CI, 9 to 52) among those with follicular lymphoma; the 10-year overall survival was 17% (95% CI, 5 to 34) and 50% (95% CI, 23 to 72), respectively. Persistent grade 2 or 3 neutropenia occurred in 2 patients (5%); no late anemia or thrombocytopenia was observed. A second primary cancer developed in 9 patients (10-year cumulative incidence, 21%). The 10-year non–relapse-related mortality was 18% (14% when deaths related to coronavirus disease 2019 were excluded). Higher CAR-transgene persistence appeared to be associated with long-term response. B-cell aplasia persisted in 44% of patients with a long-term response.

CONCLUSIONS

Among patients with heavily pretreated B-cell non-Hodgkin lymphoma, a single infusion of tisagenlecleucel led to decade-long remissions (lymphoma-free survival) in approximately one third of the patients with large B-cell lymphomas and in nearly one half of those with follicular lymphoma. (Funded by the Richard Berman Family Innovations Center in CLL and Lymphomas and others; ClinicalTrials.gov number, NCT02030834.)

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CHIMERIC ANTIGEN RECEPTOR (CAR) T-cell therapies targeting CD19 have transformed the treatment of relapsed or refractory B-cell non-Hodgkin lymphomas. Among patients with relapsed or refractory large B-cell lymphoma or follicular lymphoma, CD19-directed CAR T-cell therapies lead to durable remissions in approximately 30 to 50% of patients over a follow-up period of up to 5 years.¹⁻¹³ However, the durability of remissions beyond 5 years and the curative potential of CAR T-cell therapy remain unclear.

Long-term surveillance is essential to determine whether CAR T-cell therapies can result in lifelong remission; late recurrences after chemimmunotherapy are well recognized.¹⁴ Extended follow-up also provides an opportunity to characterize late adverse events, including non-relapse-related death and second primary cancers.^{2,15-18} Persistent cytopenias, prolonged B-cell aplasia, and hypogammaglobulinemia may increase the risk of infection for years after the infusion. The frequency and clinical consequences of these effects over an extended period remain unclear.^{2,19} Here, we report the 10-year follow-up of patients with refractory or multiply relapsed B-cell non-Hodgkin lymphoma who had been treated with CD19-directed CAR T cells in a single-center, phase 2 clinical trial,^{3,20} providing a decade-long evaluation of efficacy, toxic effects, immune reconstitution, and survival.

METHODS

STUDY DESIGN AND OVERSIGHT

We assessed 10-year outcomes in patients with relapsed or refractory large B-cell lymphoma or follicular lymphoma who had been treated at the University of Pennsylvania with CTL019 (now called tisagenlecleucel) — autologous T cells transduced by means of a lentiviral vector to express CD19-directed, 4-1BB–costimulated, chimeric antigen receptors.^{3,20} The study design, inclusion criteria, manufacturing process, treatment procedures, and initial efficacy and safety results have been published previously.^{3,20}

A total of 49 patients were enrolled; 38 patients (78%) received tisagenlecleucel and could be evaluated for efficacy and safety in this long-term study.³ Eleven patients did not receive a CAR T-cell infusion; 5 had rapid disease progression, 1 withdrew consent, and in 5, the manufacturing process did not yield the protocol-specified CAR T-cell dose.³

All the patients provided written informed consent. The institutional review board at the University of Pennsylvania approved this extended analysis, which was conducted in accordance with the principles of the Declaration of Helsinki and Good Clinical Practice guidelines. This study was open from January 2014 to September 2019 (primary study completion); the data-cutoff date for the long-term follow-up was October 1, 2025. All the authors contributed to the writing of the manuscript; no one who is not an author contributed to the writing of the manuscript. The authors vouch for the accuracy and completeness of the data and for the adherence of the study to the protocol (available with the full text of this article at NEJM.org).

TREATMENT AND ASSESSMENTS

Information about the CAR construct, lymphodepleting regimens, and protocol procedures has been published previously.^{3,20} All tisagenlecleucel products were manufactured at the University of Pennsylvania. Tumor assessments were performed according to the 2007 Revised Response Criteria for Malignant Lymphoma.²¹ Given that the last observed relapse occurred 5.4 years after the infusion, any patients who were in continuous complete remission for 5.5 years and beyond (without additional treatment or maintenance therapy after the infusion of tisagenlecleucel) were considered to have had a long-term response.

Lymphoma-free survival was defined as the time from the tisagenlecleucel infusion to lymphoma relapse or lymphoma-related death, with deaths unrelated to lymphoma censored at the time of death. Progression-free survival was defined as the time from the tisagenlecleucel infusion to lymphoma progression or death from any cause. Duration of response was measured from the date of the first response to the occurrence of relapse or death from lymphoma. Overall survival was defined as the time from the tisagenlecleucel infusion to death from any cause. Non-relapse-related death was defined as death without progression or relapse of lymphoma. Second primary cancers included all new malignant neoplasms, excluding nonmelanoma skin cancers, that were diagnosed after the tisagenlecleucel infusion. Details about the data used for mortality and cancer incidence in the general U.S. population are provided in the Supplementary Appendix (available at NEJM.org).²²⁻²⁴ Toxic effects during long-term follow-up were graded according to the

Table 1. Clinical Characteristics of the Patients with a Long-Term Response.*

Characteristic	Patients (N=12)
Median age (range) — yr	57 (49–71)
Sex — no. (%)	
Female	9 (75)
Male	3 (25)
Diagnosis — no. (%)	
Follicular lymphoma	6 (50)
Diffuse large B-cell lymphoma, not otherwise specified	2 (17)
T-cell histiocyte-rich large B-cell lymphoma	1 (8)
Transformed follicular lymphoma	3 (25)
Stage IV disease at enrollment — no. (%)	6 (50)
Median no. of previous therapies (range)	4 (2–9)
Previous autologous stem-cell transplantation — no. (%)	6 (50)
Previous allogeneic stem-cell transplantation — no. (%)	0
Lactate dehydrogenase level $\geq$ ULN at infusion — no. (%)	5 (42)

* Patients with relapsed or refractory large B-cell lymphoma or follicular lymphoma who were in continuous complete remission for 5.5 years and beyond (without additional treatment or maintenance therapy after the infusion of CTL019 [now called tisagenlecleucel]) were considered to have had a long-term response on the basis of the last observed relapse at 5.4 years. ULN denotes upper limit of the normal range.

Common Terminology Criteria for Adverse Events, version 5.0, of the National Cancer Institute.

IMMUNE MONITORING

CD3+, CD4+, and CD8+ T-cell counts, B-cell counts, and information on the use of intravenous immune globulin were abstracted from medical records. Details of correlative analyses are provided in the Supplementary Appendix.

STATISTICAL ANALYSIS

Continuous variables are summarized as medians and ranges, and categorical variables as counts and percentages. We used Kaplan–Meier estimates for lymphoma-free survival, progression-free survival, response duration, and overall survival, with 95% confidence intervals computed by Greenwood's formula (log–log transformation). Patients without an event had their data censored at the date of the last assessment. The incidence of non–relapse-related death and second primary cancer was estimated with Aalen–Johansen methods, with lymphoma progression treated as a

competing risk for non–relapse-related death and with death from any cause treated as a competing risk for second primary cancer. Missing data were handled by a complete-case analysis without imputation to maintain the primary integrity of the long-term clinical observations.

RESULTS

LONG-TERM EFFICACY

At a median follow-up of 10.1 years (range, 7.9 to 11.5), approximately one third of the patients remained in remission. The last lymphoma relapse occurred 5.4 years after the tisagenlecleucel infusion in a patient with primary refractory large B-cell lymphoma. At the time of relapse in this patient, pathological testing showed CD19-positive large B-cell lymphoma clonally related to the primary lymphoma with abundant CD3+ T-cell infiltration (CD19 H-score, 240; a semi-quantitative, numerical value used in pathological assessments to evaluate protein expression in tissue samples, typically ranging from 0 to 300, on the basis of staining intensity [weak, moderate, or strong]); this patient's CAR-transgene level in peripheral blood was undetectable beyond 2 years after the tisagenlecleucel infusion. Among patients with follicular lymphoma, no relapses were observed beyond 2.7 years. A total of 12 of 38 patients (32%) met the criteria for a long-term response. The characteristics of these 12 patients are shown in Table 1.

The 10-year lymphoma-free survival was 38% (95% confidence interval [CI], 22 to 53) among all the patients who had received the infusion and could be evaluated (Fig. S1A in the Supplementary Appendix). At 10 years, lymphoma-free survival was 32% (95% CI, 14 to 51) among patients with large B-cell lymphoma and 47% (95% CI, 20 to 70) among those with follicular lymphoma (Fig. 1A and 1B). Deaths during remission led to 10-year progression-free survival estimates of 21% (95% CI, 10 to 35) among all patients who had received the infusion, 17% (95% CI, 5 to 34) among patients with large B-cell lymphoma, and 29% (95% CI, 9 to 52) among those with follicular lymphoma (Fig. 1 and Fig. S1B) and to 10-year overall survival estimates of 29% (95% CI, 16 to 44), 17% (95% CI, 5 to 34), and 50% (95% CI, 23 to 72), respectively (Fig. 1 and

Fig. S1C). Among patients with a response, the Kaplan–Meier estimate of a continued response at 10 years was 57% (95% CI, 35 to 74) among all the patients, 54% (95% CI, 25 to 77) among those with large B-cell lymphoma, and 60% (95% CI, 25 to 83) among those with follicular lymphoma (Fig. 1 and Fig. S1D).

LONG-TERM SAFETY

Long-term hematologic recovery was assessed; chronic anemia or thrombocytopenia was not observed (Fig. 2A), and 36 of 38 patients had stable neutrophil recovery. Two patients had ongoing grade 2 or 3 neutropenia at the last follow-up, with the neutropenia in one patient being related to preexisting autoimmune neutropenia (granulocyte-colony stimulating factor–responsive) and the neutropenia in the other patient being related to clonal cytopenia of undetermined significance without morphologic or karyotypic abnormalities in bone marrow (1712 tisagenlecleucel copies per microgram of DNA in peripheral blood at the diagnosis of clonal cytopenia of undetermined significance) (Figs. S2 and S3). *DNMT3A* and *TET2* mutations, lesions known to be associated with clonal hematopoiesis, were detected in the peripheral blood of the latter patient.²⁵

A second primary cancer developed in nine patients after the infusion, with a median time to the diagnosis of second cancer of 4.7 years (range, 0.7 to 10.8). Second cancers included therapy-related acute myeloid leukemia (in three patients [8%], all of whom had characteristic therapy-associated genetic mutations), prostate cancer (in two [5%]), non–small-cell lung carcinoma (in two [5%], both of whom were smokers), mycosis fungoides (in one [3%], with a CAR transgene–negative tumor), and melanoma (in one [3%]). Data regarding the clinical characteristics of these patients and the time to the onset of second cancers are provided in Figure S4 and Table S1.

In this study, the 10-year cumulative incidence of second primary cancer was 21% (95% CI, 8 to 34) (Fig. 2B). In a U.S.-matched population (Surveillance, Epidemiology, and End Results [SEER] database), the 10-year cancer risk is 11%.²²

The 10-year cumulative incidence of non–relapse-related death was 18% (95% CI, 6 to 31), and the cumulative incidence was 14% (95% CI,

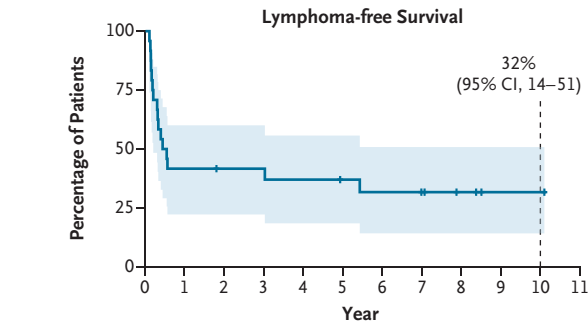
2 to 24) when deaths related to coronavirus disease 2019 (Covid-19) were excluded (Fig. 2C and Fig. S5A). Causes of non–relapse-related death included second cancers (in three patients [8%]), Covid-19 (in two [5%]), cardiac arrest related to anthracycline-induced cardiomyopathy (in one [3%]), and respiratory failure in the context of nonmalignant cerebellar dysfunction (in one [3%]) (Table S2). The median time to non–relapse-related death was 6.2 years (range, 0.6 to 7.5). The 10-year mortality in an age- and sex-matched U.S. population is 10%, whereas the 10-year non–cancer-related mortality in a matched lymphoma population is 20% (Fig. 2C and Fig. S5B, S5C, and S5D).^{23,24}

CAR T-CELL PERSISTENCE AND IMMUNE RECONSTITUTION

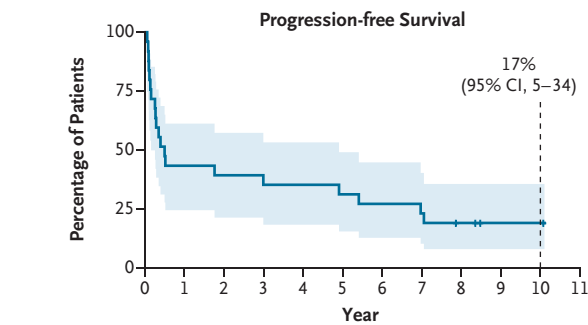
CAR-transgene DNA levels from day 28 through the first 2 years after the infusion appeared to be higher among patients with a long-term response than among those without a long-term response (Fig. S6 and Table S3). At the last follow-up, 7 of 12 patients (58%) with a long-term response had CD3+ T-cell counts within the normal range, 11 of 12 (92%) had CD8+ counts within the normal range, and 6 of 12 (50%) had CD4+ T-cell counts within the normal range. All the patients recovered CD4+ T-cell counts above 200 cells per microliter over time (Fig. S7A, S7B, and S7C). By 10 years of follow-up, 6 patients with a long-term response who could be evaluated had broad immune reconstitution, with a numerically higher terminally differentiated effector memory T-cell count (TEMRA fraction) than 131 persons in a healthy matched cohort,²⁶ a finding that suggests a potential difference in T-cell subset reconstitution after tisagenlecleucel infusion (Fig. S7D and Tables S4 and S5).

As of the data-cutoff date, 4 of 9 patients (44%) who could be evaluated had persistent B-cell aplasia (Fig. S8A). A total of 7 of 12 patients (58%) received intravenous immune globulin, and as of the last follow-up, 5 patients (42%) were continuing to receive intravenous immune globulin. Reconstituted B-cell subsets (in 4 patients) were broadly similar to those in 144 adults in a healthy matched cohort (Fig. S8B and Table S6).^{27–29}

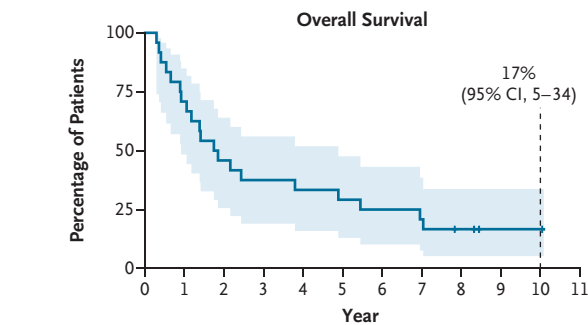
Beyond 5 years, infectious events were reported

A Patients with Large B-Cell Lymphoma

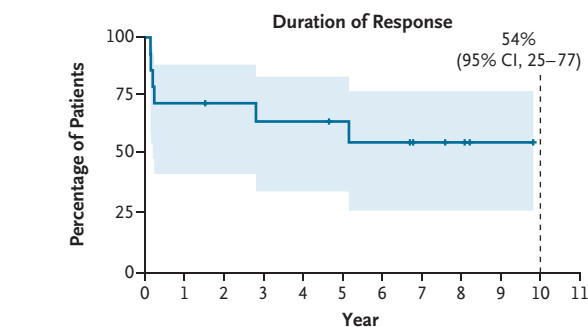
No. at Risk 24 10 9 8 8 7 6 5 3 1 1 0



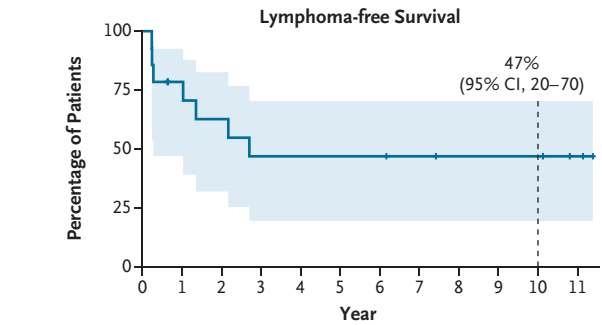
No. at Risk 24 10 9 8 8 7 6 5 3 1 1 0



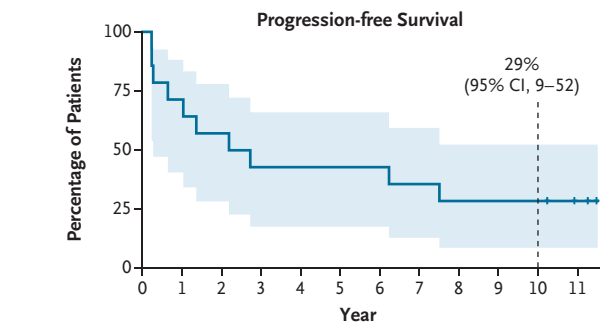
No. at Risk 24 17 11 9 8 7 6 5 3 1 1 0



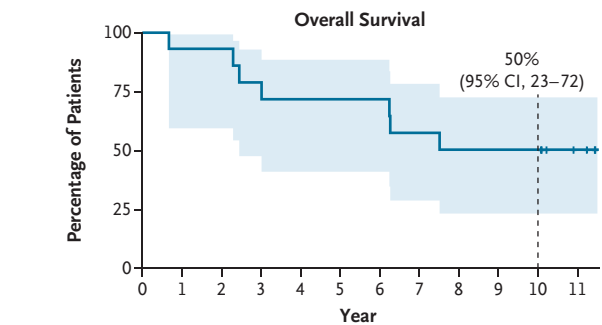
No. at Risk 14 10 9 8 8 7 6 4 3 1 0 0

B Patients with Follicular Lymphoma

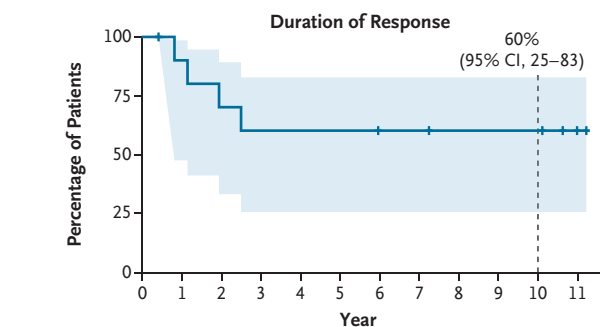
No. at Risk 14 10 8 6 6 6 6 5 4 4 4 2



No. at Risk 14 10 8 6 6 6 6 5 4 4 4 2



No. at Risk 14 13 13 10 10 10 10 8 7 7 7 3



No. at Risk 11 9 7 6 6 6 5 5 4 4 4 2

Figure 1 (facing page). Long-Term Efficacy.

Shown are Kaplan–Meier survival curves of lymphoma-free survival, progression-free survival, overall survival, and duration of response among patients with large B-cell lymphomas (Panel A) and among patients with follicular lymphoma (Panel B). Tick marks indicate censored data, and shaded areas 95% confidence intervals.

in 8 of 12 patients (67%) with a long-term response; 3 of 14 events (21%) were severe. Covid-19 developed in 3 patients with a long-term response; 2 unvaccinated patients were hospitalized and died. Humoral immune responses to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) messenger RNA vaccination that had been administered 45 to 82 months after the infusion of CAR T cells were assessed in 4 patients with a long-term response. Three of these 4 patients had positive SARS-CoV-2 receptor binding domain IgG.³⁰ No late opportunistic infections were observed (Table S7).

DISCUSSION

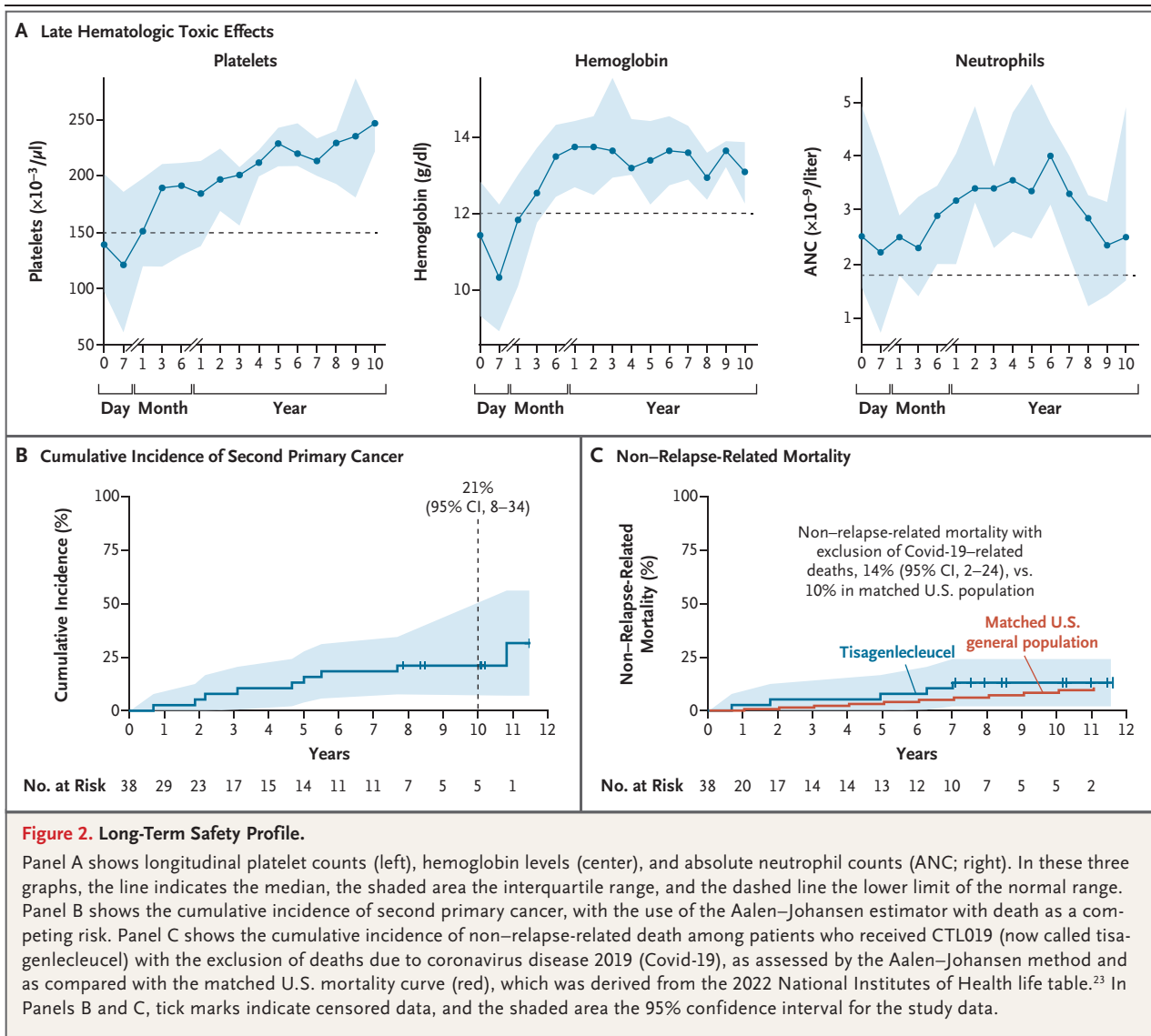
In this 10-year follow-up study of CD19-directed CAR T-cell therapy for relapsed or refractory B-cell non-Hodgkin lymphoma, we found that most treatment failures or relapses occurred within the first year after the infusion, a result that is consistent with previous reports.^{4-9,20,31,32} No relapses were seen beyond 5.4 years, and approximately one third of all the patients remained in continuous remission at 10 years, with lymphoma-free survival at 10 years of 32% among patients with large B-cell lymphoma and 47% among those with follicular lymphoma. These results extend those of previous analyses, which showed 35 to 55% progression-free survival, with a plateauing of the survival curves beyond 1 year.^{2,3,13} These 10-year data confirm that early survival-curve plateaus may predict durable remissions, which is unlike the results seen with chemoimmunotherapy or bispecific antibody therapies, which are more frequently characterized by late relapses, especially in patients with follicular lymphoma.^{14,33,34} We observed no relapses beyond 2.7 years among patients with follicular lymphoma, which suggests that the findings of long-term analyses of the ELARA and ZUMA-5 trials may potentially extend to the 10-year bench-

mark.^{12,35} Our findings support the hypothesis that patients with a long-term response to CAR T-cell therapy may be cured — including patients with heavily pretreated large B-cell lymphomas or follicular lymphoma, who are generally considered to have incurable disease.

The 10-year cumulative incidence of second primary cancer was 21%. No cases of CAR T-cell-related lymphomas were observed. The observed incidence of second cancer exceeds the 6% incidence that was reported by Tix et al. in a large meta-analysis,¹⁷ a finding that possibly reflects the longer observation period or extensive pretreatment (or both) in our study cohort. Although the 10-year incidence of second primary cancer that we observed is numerically higher than the cancer risk in a matched U.S. population, it aligns closely with data from a Danish registry of patients with lymphoma treated with high-dose chemotherapy and autologous transplantation, a finding that underscores the hypothesis that previous exposure to chemotherapy may be a primary driver or at least a cofactor of long-term risk of second primary cancers.^{18,22} If early exposure to cytotoxic chemotherapy is playing a role in the incidence of second cancers, an argument could be made for moving CAR T-cell therapy earlier in the line of therapies to reduce the risk of second cancer by reducing the cumulative exposure to genotoxic chemotherapy.³⁶

Earlier studies linked the magnitude of early CAR T-cell expansion in blood with durable remissions.^{2,37-40} In our study, patients with long-term remissions appeared to have higher CAR T-cell levels over the first 2 years after the infusion than did patients without long-term remission. Thus, the magnitude of CAR T-cell persistence over time, rather than early peak expansion — a key driver of durable responses in CD28-costimulated CAR T cells — may be critical for long-term outcome with 4-1BB-based CAR T-cell products.^{2,40}

Delayed recovery of T cells and persistent B-cell aplasia, which are risk factors for infection after the receipt of CAR T-cell therapy, represent a major cause of non-relapse-related death.^{2,16} In the present study, 58% of the patients with a long-term response had normalization of T-cell counts at the last follow-up, with reconstitution of CD4+ and CD8+ T cells in 50% and 92% of patients, respectively. Luan et al.⁴¹



observed that nearly half of patients had CD4+ counts below 200 per microliter after the receipt of CAR T-cell therapy, a threshold associated with a risk of opportunistic infections. However, among the patients with a long-term response in our study, no patient had counts below this threshold, a finding that suggests that CD4+ recovery may eventually reach a threshold sufficient to mitigate the risk of infection.

Nearly half the patients with a long-term response had persistent B-cell aplasia, and 42% were continuing to receive intravenous immune globulin at 10 years. These findings are in line

with previous reports^{2,3} and support the use of infection prophylaxis and immune globulin supplementation as indicated by immunologic recovery.

This study is limited by its single-center design, modest sample size, and the lack of systematically collected correlative data at late time points, which precludes a conclusive, detailed characterization of immune-cell subset reconstitution and mechanisms underlying specific second primary cancers. Nevertheless, the study establishes that CD19-directed CAR T-cell therapy can lead to exceptionally long remissions and

possibly cure some patients with relapsed or refractory non-Hodgkin lymphoma.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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